Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **Python AI platforms**, **full-stack workflows**, and **Kubernetes** infrastructure with PostgreSQL, React, document retrieval, and production LLM systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Tripled throughput across 1K+ AI jobs by restructuring **fault-tolerant FastAPI orchestration** with explicit job states, background workers, failure isolation, recoverable processing, and observable production workflows under concurrent load.
- Reduced dashboard latency from 1.2s to 350ms by optimizing **PostgreSQL schemas**, indexed queries, row-level isolation, and access patterns for a multi-tenant AI platform supporting 500+ users reliably.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround 60% by delivering a **React and Flask platform** for 150+ instructors and 10K+ daily grading requests, integrating multimodal LLMs, OCR, REST APIs, and AWS RDS.
- Processed 1K+ files per batch using **Python multithreading**, validating PDF, DOCX, and ZIP inputs while isolating failures and preserving responsive workflows during concurrent model execution at scale.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments at scale.
- Increased data-pipeline efficiency 68% by redesigning **AWS Glue and S3 ETL** workflows, automating JSON, CSV, and Parquet transformations for reliable API consumption and analytics across production environments.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **real-time microservice platform** using React, JavaScript, Node.js, WebSockets, REST APIs, and Azure SQL across distributed business workflows and customer-facing operational requirements.
- Reduced resource wastage by 42% through **data synchronization and testing**, integrating automated suites, Docker/Jenkins delivery, Azure Monitor, Sentry, and Kibana across modular production services reliably at scale.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **Flask APIs** for a cross-platform service-booking product covering time slots, pricing, coupons, payments, refunds, disputes, and real-time Firebase synchronization across 50K+ booking records with responsive workflows.
- Reduced manual refund handling approximately 70% by integrating **payment webhooks**, status updates, dispute workflows, and sub-200ms booking APIs across mobile and web experiences for operational teams reliably.

Skills

Languages: Python, JavaScript, TypeScript, SQL, Java, C++, Bash
AI & Search: LLMs, Document Retrieval, Semantic Search, FAISS, Pinecone, LLaMA-2, Transformers, OCR, Prompt Engineering
Backend & Data: FastAPI, Flask, REST APIs, PostgreSQL, AWS RDS, Azure SQL, MongoDB, Redis, Microservices, Data Modeling
Frontend: React, Next.js, Responsive Design, WebSockets, TailwindCSS, HTML, CSS
Cloud & Engineering: Kubernetes, Docker, AWS, Azure, GCP, CI/CD, GitHub Actions, Jenkins, Git, PyTest, Unit Testing, Integration Testing

Projects

Big-Brain - Semantic Document Retrieval | Live

- Built a secure **semantic document retrieval** platform over 50K+ documents using LLaMA-2, FAISS, Pinecone, PostgreSQL, and REST APIs, enabling contextual Q&A, summarization, and sub-second search workflows reliably.

Private AI Review Workbench - Design Concept

- Designed an **air-gapped AI review** workbench concept using Python, FastAPI, PostgreSQL, local model inference, Kubernetes manifests, role-based access, audit trails, and explainable evidence-linked document classifications for legal workflows.

FinNexus | Live

- Built a responsive **full-stack platform** supporting 50K+ users and 1M+ transactions using Next.js, TypeScript, Kubernetes, MongoDB, WebSockets, secure APIs, real-time synchronization, filtering, and CI/CD deployment workflows reliably.